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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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EXAMINER	
NGUYEN, THANG H	

ART UNIT	PAPER NUMBER
2831	

NOTIFICATION DATE	DELIVERY MODE
04/24/2023	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 17/151,277	Applicant(s) BLACKMON et al.	
	Examiner THANG H NGUYEN	Art Unit 2831	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 7/22/22.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-20 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-6 and 9-20 is/are rejected.

8) ☑ Claim(s) 7-8 is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 1/18/21 is/are: a)☐ accepted or b)☑ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Drawings

The drawings are objected to under 37 CFR 1.83(a) because they fail to show "**second upstream grooves**" as described in the specification. Any structural detail that is essential for a proper understanding of the disclosed invention should be shown in the drawing. MPEP § 608.02(d). Corrected drawing sheets in compliance with 37 CFR 1.121(d) are required in reply to the Office action to avoid abandonment of the application. Any amended replacement drawing sheet should include all of the figures appearing on the immediate prior version of the sheet, even if only one figure is being amended. The figure or figure number of an amended drawing should not be labeled as "amended." If a drawing figure is to be canceled, the appropriate figure must be removed from the replacement sheet, and where necessary, the remaining figures must be renumbered and appropriate changes made to the brief description of the several views of the drawings for consistency. Additional replacement sheets may be necessary to show the renumbering of the remaining figures. Each drawing sheet submitted after the filing date of an application must be labeled in the top margin as either "Replacement Sheet" or "New Sheet" pursuant to 37 CFR 1.121(d). If the changes are not accepted by the examiner, the applicant will be notified and informed of any required corrective action in the next Office action. The objection to the drawings will not be held in abeyance.

Claim Rejections - 35 USC § 102

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

> A person shall be entitled to a patent unless –
>
> (a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale, or otherwise available to the public before the effective filing date of the claimed invention.

Claim(s) 1 as best understood and is/are rejected under 35 U.S.C. 102(a)(1) as being anticipated by Blandino et al. (US 2017/0055584).

As per claim 1: Blandino discloses a capsule (cartridge) 1 for an aerosol generating device 1000 comprising: a housing 10 defining inlet openings 14, outlet openings 16, and a chamber 18 between the inlet openings 14 and the outlet openings 16 (see Para. [0078]; wherein one or both of the end member **14** and the end closure **16** may be openable, puncturable or removable from the article **1** before use, so as to enable air flow through the cavity **18** and thus through the smokable material **30**. However, in some embodiments, one or both of the end member **14** and the end closure **16** may comprise an air-permeable membrane or cover for admitting air to pass between the cavity **18** and an exterior of the article **1)**, the chamber 18 having a longest dimension extending from at least one of the inlet openings 14 to a corresponding one of the outlet openings 16 (as shown in figs. 1-2); an aerosol-forming substrate 30 within the chamber 18 of the housing 10; and a heater 20 extending into the housing 10, the heater including a 1st end section, an intermediate section, and a 2nd end section (see figs. 1-2; wherein the 1st end section is adjacent to the inlet openings, and the 2nd section is adjacent to the outlet openings 16, and the intermediate section 24), the intermediate section 24 being disposed within the aerosol-forming substrate 30 in the chamber 18.

Claim(s) 1 as best understood and is/are also rejected under 35 U.S.C. 102(a)(1) as being anticipated by Hepworth et al. (US 2020/0390149).

As per claim 1: Hepworth discloses a capsule 4 for an aerosol-generating device 1, comprising: a housing 60 defining inlet openings, outlet openings (see fig. 1; wherein the both end of the capsule/cartridge are having openings 68), and a chamber 66 between the inlet openings and the outlet openings, the chamber 66 having a longest dimension extending from at least one of the inlet openings to

a corresponding one of the outlet openings (as shown in fig. 1); an aerosol-forming substrate 63 within the chamber 66 of the housing 60; and a heater 70 extending into the housing 60, the heater 70 including a first end section, an intermediate section, and a second end section, the intermediate section being disposed within the aerosol-forming substrate in the chamber 66 (see fig. 1; wherein the 1st end section is adjacent to the inlet openings 68, and the 2nd section is adjacent to the outlet openings 68 (opposite of the other openings 68), and the intermediate section is located in the middle of the housing).

Claim(s) 1, 18-19 as best understood and is/are also rejected under 35 U.S.C. 102(a)(1) as being anticipated by Reevell (US 2018/0263286).

As per claim 1: Reevell discloses capsule (31,32,33) for an aerosol-generating device (as shown in figs.12-16), comprising: a housing defining inlet openings (not shown, see figs. 6, and 9-10; wherein each capsule is having an inlet and outlet), outlet openings (210,220,230), and a chamber between the inlet openings and the outlet openings (see figs. 9-10; Para. [0135]; wherein each consumable are different types of consumables to contain reservoir, article, and/or solid tobacco material containing article), the chamber having a longest dimension extending from at least one of the inlet openings to a corresponding one of the outlet openings (210,220,230) (see figs. 6, 9-11); an aerosol-forming substrate within the chamber of the housing (see Para. [0135]); and a heater 66 extending into the housing (see fig. 10), the heater 66 including a first end section, an intermediate section, and a second end section, the intermediate section being disposed within the aerosol-forming substrate in the chamber (see fig. 10; wherein the 1st section is adjacent to the inlet portion, the 2nd portion section is adjacent to the outlet portion, and the middle portion is intermediate portion).

As per claim 18: Reevell discloses an aerosol-generating device (as shown in figs. 1, 4), comprising: a capsule (31/32/33) including a housing (as shown in fig. 6; Para. [0135], wherein each individual capsule is aerosol generating article comprising aerosol forming substrate), an aerosol-forming substrate, and a heater (as shown in figs. 9-10, elements 60/66), the housing defining inlet openings, outlet openings (210/220/230), and a chamber between the inlet openings

and the outlet openings (see figs. 9-11; wherein there is an inlet/outlet opening (as shown following an arrow) on each individual capsule, and a chamber to store the aerosol forming substrate) a mouthpiece (21/22/23) configured to engage with the capsule 31/32/33 so as to be in fluidic communication with the chamber via the outlet openings (out to the user); and a device body 1 configured to receive and retain the capsule 31/32/33 and the mouthpiece (21/22/23) (see figs. 1, 4) the device body 1 including a power source 16 configured to supply an electric current to the heater to heat the aerosol-forming substrate (to be able to generate aerosol to the user).

As per claim 19: Reevell discloses the aerosol-generating device, wherein the device body 1 includes a door 20 configured to open to permit an insertion of the capsule (31/32/33) and the mouthpiece (21/22/23) and configured to close to retain the capsule (31/32/33) and the mouthpiece (21/22/23) (as shown in figs. 1-2, 4).

Claim(s) 1-5,12-13,15, and 17-18 as best understood and is/are also rejected under 35 U.S.C. 102(a)(1) as being anticipated by Reevell (US 2018/0214645).

As per claim 1: Reevell discloses a capsule (41/42) for an aerosol generating device 1 (as shown in fig. 31) comprising: a housing (not shown, see fig. 9; wherein shown the 1st and 2nd type consumable 41/42) defining inlet openings, outlet openings, and a chamber between the inlet openings and the outlet openings (see figs. 10-13; wherein there is an inlet to the consumable 41/42, and an outlet from consumable 41/42, and there is a cavity/chamber in between the inlet and outlet openings), the chamber having a longest dimension extending from at least one of the inlet openings to a corresponding one of the outlet openings (as shown in figs. 10-12; wherein the dimension of the chamber go from one end to other end of the consumable 41/42) an aerosol-forming substrate 410 within the chamber (wherein the consumable 41/42 are containing tobacco material) of the housing (as shown in figs. 10-13); and a heater 66 extending into the housing, the heater 66 including a first end section, an intermediate section, and a second end section (as shown in figs. 6, 12, 21, and 29; wherein the heater 66 has 1st portion at one end opposite to 2nd portion at the other end, and in between is the middle section (intermediate section), the intermediate section being disposed within the aerosol-forming substrate in the chamber (as shown in

figs. 6, 12, 21, and 29; wherein the middle section of the heater 66 is located within the aerosol substrate (1st and 2nd type of consumable 41/42).

As per claims 2-5; 12-13, 15, and 17-18: Reevell discloses the capsule (41/42) for an aerosol generating device 1 (as shown in fig. 31) wherein the housing has a longitudinal axis, and the longest dimension of the chamber extends along the longitudinal axis of the housing (see figs. 9, 18, 24, 32-33, 36; wherein the longitudinal shape housing which contained the aerosol forming substrate/consumable material); and wherein the housing includes a body portion and a base portion, the body portion including a proximal end and a distal end, the base portion configured to engage with the distal end of the body portion (see fig. 36, wherein the consumable insert 44 is including the base portion 446 connected to the cylindrical portion 440, and optionally also covers the flaps 441); and wherein the base portion of the housing defines an air channel (as indicated in the arrows from inlet to outlet of the capsule/consumable) in fluidic communication with the chamber via the inlet openings (see figs. 12-13, 21-22; in order to generated aerosol to the user); and wherein the body portion of the housing defines the inlet openings, the outlet openings, and the chamber (see figs. 9, 18-20; wherein each of the capsule/consumable is containing a reservoir with material/article/substance within and having inlet/outlet opening in order for the air to pass through and generate aerosol as should be); wherein the aerosol forming substrate 410 includes a plant material, and the plant material includes tobacco (see Para. [0111, 0123]); and wherein the intermediate section of the heater 66 is between the inlet opening and the outlet openings (see figs. 12-13, 21-22); and wherein at least one annular member engaged with the housing (see figs. 15-17, 19-20, and 23-26).

As per claim 18: Reevell discloses an aerosol-generating device 1, comprising: a capsule 41/42 including a housing (not shown, see fig. 9; wherein shown the 1st and 2nd type consumable 41/42), an aerosol-forming substrate 410, and a heater 66, the housing defining inlet openings, outlet openings, and a chamber between the inlet openings and the outlet openings (see figs. 10-13; wherein there is an inlet to the consumable 41/42, and an outlet from consumable 41/42, and there is a cavity/chamber in between the inlet and outlet openings)); a mouthpiece 20 (see fig. 31) configured to engage with the capsule

(41/42)(see fig.31, Para. [0179, 0194) so as to be in fluidic communication with the chamber via the outlet openings (wherein the aerosol on its way to the user); and a device body 2 configured to receive and retain the capsule 41/42 and the mouthpiece 20, the device body 2 including a power source 22 configured to supply an electric current to the heater 66 to heat the aerosol-forming substrate 410 (see Para. [0133-0134]).

Claim Rejections - 35 USC § 103

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

Claim(s) 6, 9-11, 14, 16 as best understood and is/are rejected under 35 U.S.C. 103 as being unpatentable over Reevell (US 2018/0214645).

As per claim 6, 9-10: Reevell discloses the capsule (41/42) for an aerosol generating device 1 (as shown in fig. 31). However, Reevell does not explicitly disclose wherein the body portion of the housing includes a first body component and a second body component; and wherein the first body component and the second body component are identical and complementary structures; and wherein the first body component and the second body component are configured to clamp onto the base portion.

On the other hand, Reevell teaches the capsules/consumables (41, 42) with a one integral piece housing, and a base (as shown in fig. 36) is attached to the consumable/capsule. However, a person having ordinary skill in the art would know that having a body portion with 2 pieces component instead of an integral piece can only deal with making separable because such a modification still would not change the function of the consumable/capsule and the aerosol generating device after all, but as the

manufacture desire design to adopt the structure. Further, having a base clamps to the 1st and 2nd body component can only deal with one of many different methods of connecting parts together such as by glue, solder, epoxy etc. in order to provide a stable, reliable and secure structure between components. Since, it has been held that constructing a formerly integral structure in various elements involves only routine skill in the art. *Nerwin v. Erlicnrnan,* 168 USPQ 177, 179.

Therefore, it would have been obvious to one having ordinary skill in the art before the effective filing date of the claim invention to further modify the capsule for an aerosol generating device of Reevell by having the body portion of the housing includes a first body component and a second body component; and wherein the first body component and the second body component are identical and complementary structures; and wherein the first body component and the second body component are configured to clamp onto the base portion as taught by the instant invention as the manufacture desire design as well as to further provide a stable, reliable and secure structure between components.

As per claim 11: Reevell discloses the capsule (41/42) for an aerosol generating device 1 (as shown in fig. 31), and the aerosol forming substrate 410, and the heater 66. However, Reevell does not explicitly disclose wherein the aerosol-forming substrate includes a first consolidated form and a second consolidated form disposed on opposite sides of the intermediate section of the heater.

On the other hand, Reevell teaches each consumable/capsule (41/42) contains a storage portion which accommodated material/article/substance and the heater within in order to generate aerosol to the user; and when the flaps serve as liquid aerosol-forming substrate reservoir, **the consumable insert is a hybrid consumable providing two different substrates**. In an embodiment, a tobacco substrate is combined with a liquid substrate, for example providing at least one of nicotine and flavor (see Para. [0123]). However, a person having ordinary skill in the art would know that having the aerosol forming substrate that has 2 different consolidated form in opposite sides of the intermediate heater can only deal with to make adjustable because such modification still would not change the function of the capsule/consumable/cartridge in the device but to further enhance more options to the user to choose from in which consolidated form/articles/substances the user would like to use, and to accommodate

different flavors in different compartments of the substrate. Since, it has been held that the provision of adjustability, where needed, involves routine skill in the art . In re Stevens, *101 USPQ 284 (CCPA* 1954).

Therefore, it would have been obvious to one having ordinary skill in the art before the effective filing date of the claim invention to further modify the capsule for an aerosol generating device of Reevell by having the aerosol-forming substrate includes a first consolidated form and a second consolidated form disposed on opposite sides of the intermediate section of the heater as taught by the instant invention to further provide more options for the user to choose from as well as to enhance the device in the competitive market nowadays.

As per claim 14: Reevell discloses the capsule (41/42) for an aerosol generating device 1 (as shown in fig. 31), and the heater (as shown in fig. 33). However, Reevell does not explicitly disclose wherein the heater is embedded through the housing such that segments of the 1st end and 2nd end section are external to the housing.

On the other hand, Reevell discloses the housing insert 30 with inductive heating 65 on the external wall of the housing portion insert 30 an induction coil is split axially and radially and forms an arrangement of induction coils 65. Two split coils 65 are each arranged on the cylindrical portion 31 and on the cuboid portion 32. In an embodiment, the coil arrangement is symmetric on both sides of the housing portion insert 30. In an embodiment, the individual induction coils 65 may be activated at appropriate frequencies in a continuous or pulsed power mode to inductively power any type of inductively activatable consumable accommodated in the receiving chamber 3 (see Para. [0186]); or other embodiment that the heater 66 is embedded in the housing of the capsule/consumable (as shown in figs. 12-13, 21-22, 29-30). However, a person having ordinary skill in the art would know that having the heater as recited above can only deal with location of part, because such modification still would not change the function of the heater as well as the aerosol generating device after all, but to further enhance the effective of the heater to the substance/article/material/liquid in the capsule/consumable/cartridge in order to generate the aerosol to the user as best. Since, it has been held that rearranging parts of an invention involves only routine skill in the art. *In re Japikse,* 86 USPQ 70.

Therefore, it would have been obvious to one having ordinary skill in the art before the effective filing date of the claim invention to further modify the capsule for an aerosol generating device of Reevell by having the heater is embedded through the housing such that segments of the 1st end and 2nd end section are external to the housing as taught by the instant invention to further enhance a stability, reliability and effective heating to the substance/article/material/liquid in the capsule/consumable/cartridge in order to generate the aerosol to the user as best.

As per claim 16: Reevell discloses the capsule (41/42) for an aerosol generating device 1 (as shown in fig. 31), and the heater (as shown in fig. 33). However, Reevell does not explicitly disclose wherein the intermediate section of the heater has a planar and winding form.

On the other hand, Reevell mentioned the heater as an elongated arranged in the consumable. However, a person having ordinary skill in the art would know that having the intermediate section of the heater has a planar and winding form can only deal with change in shape because such modification still would not change the function of the heater of the aerosol generating device after all but to further adapt to the structure or design of the manufacture desire according to the device to further enhance the effective of the heater to the aerosol generating device and in order to generate the aerosol to the user as best. Since, a change in shape is generally recognizing as being within the level of ordinary skill in the art. In re Dailey, 149 USPQ 47 (CCPA 1976).

Therefore, it would have been obvious to one having ordinary skill in the art before the effective filing date of the claim invention to further modify the capsule for an aerosol generating device of Reevell by having the intermediate section of the heater has a planar and winding form as taught by the instant invention to enhance the effective of the heater to the aerosol generating device and in order to generate the aerosol to the user as best.

As per claims 19-20: Reevell discloses the aerosol-generating device 1, with the capsule/consumable insert (as shown in figs. 32-33), and the mouthpiece 20 includes a mouth end and an opposing capsule end (as shown in fig. 31). However, Reevell does not explicitly disclose wherein the device body includes a door configured to open to permit an insertion of the capsule and the mouthpiece

and configured to close to retain the capsule and the mouthpiece; and the capsule end being larger than the mouth end and configured to prevent a disengagement of the mouthpiece from the capsule when the door of the device body is closed.

On the other hand, Reevell discloses wherein a device with a device housing 2 with an integrated mouthpiece 20. The device housing 2 includes a hinged housing portion 24. The receiving chamber 3 for receiving a first-type consumable or a second type consumable is arranged in the hinged housing portion 24. The hinged housing portion 24 may be tilted along a hinge (not shown) in the device housing. In a closed position of the hinged housing portion 24, the housing portion 24 is aligned with the entire housing 2 and the receiving chamber 3 is not accessible. In an open configuration of the hinged housing portion 24, the hinged housing portion 24 is tilted out of the plane of the device housing 2 and the receiving chamber 3 is accessible, the receiving chamber 3 has the form of a hollow cylinder 31 combined with a hollow cuboid 32 (as shown in fig. 31, Para. [0179]) in order to provide the user convenience accessible to replace the capsule/consumable each time when needs it. Although, Reevell does not exactly disclose the exactly structure above. However, a person having ordinary skill in the art would know that it would have been an obvious matter of design choice, because such modification still would not change the function of the capsule as well as the mouthpiece being as recited above. Further, since applicant has not disclosed the special structure above would solved any stated problem or is for any particular purpose and it appear that the invention would perform equally well with the prior art of record.

Therefore, it would have been obvious to one having ordinary skill in the art before the effective filing date of the claim invention to further modify the aerosol generating device of Reevell by having the device body includes a door configured to open to permit an insertion of the capsule and the mouthpiece and configured to close to retain the capsule and the mouthpiece; and the capsule end being larger than the mouth end and configured to prevent a disengagement of the mouthpiece from the capsule when the door of the device body is closed as taught by the instant invention to further provide easy accessible to replace the capsule as well as the convenience for the user to remove/replace the capsule if need it. Furthermore, although the prior art of record does not exactly describe the structure as recited from the instant invention, but it appears to perform equally well also.

Allowable Subject Matter

Claims 7-8 are objected to as being dependent upon a rejected base claim.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to THANG H NGUYEN whose telephone number is (571)270-0288. The examiner can normally be reached 9am-5pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Abdullah Riyami can be reached on 571-270-3119. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/T.H.N/
Examiner, Art Unit 2831
/ABDULLAH A RIYAMI/
Supervisory Patent Examiner, Art Unit 2831